

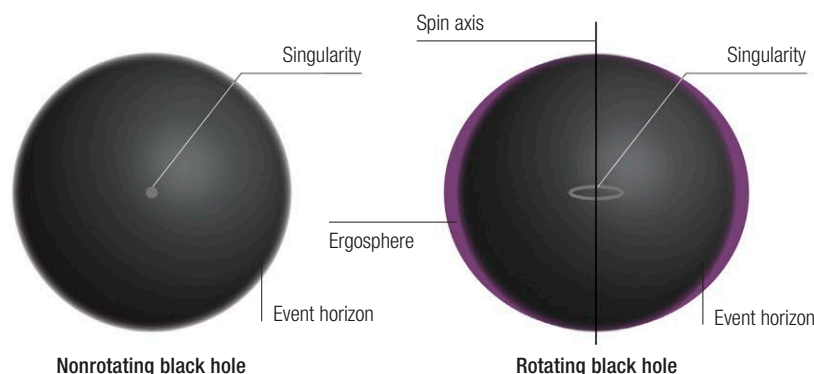
BLACK HOLES

A BLACK HOLE IS ONE OF THE STRANGEST OBJECTS IN THE UNIVERSE—A REGION OF SPACE WHERE MATTER HAS BEEN SQUEEZED INTO A MINUSCULE POINT OR RING OF INFINITE DENSITY, CALLED A SINGULARITY.

In a spherical region around the singularity, the gravitational pull toward the center is so strong that nothing, not even light, can escape. The boundary of the region of no escape is called the event horizon, and anything passing inward through this boundary can never return. There are two main types of black hole. Stellar black holes form from the collapse of the cores of supergiant stars that have exploded as supernovas at the ends of their lives. Supermassive black holes are much bigger and are thought to exist at the centers of most galaxies.

Detecting black holes

Because it emits no light, a black hole cannot be observed or imaged directly. However, some black holes can be detected from their strong gravity, which attracts other matter. These black holes may have disk-shaped collections of gas and dust around them that are spiraling into the black hole, at the same time throwing off vast amounts of X-rays or other radiation. The easiest ones to detect are those that produce jets of high-energy particles from their poles.



△ Nonrotating and rotating black holes

Black holes fall into rotating and nonrotating types—astronomers think that most rotate. In a nonrotating black hole, the singularity is a point of infinite density at the center of the hole, whereas in the rotating variety, the singularity is ring-shaped. In both types, the event horizon—the boundary of the region of no escape—forms the surface of a sphere. However, around a rotating black hole's event horizon is an additional region, the ergosphere. Anything entering this is dragged around by the black hole's spin.

▷ Gravitational light bending

A black hole's gravity is so strong that it warps nearby spacetime (see p.73) and bends the paths of passing light rays. Shown here are the paths of four, originally parallel, light rays traveling near a black hole. The first two have their paths radically altered and the third ray ends up circling the black hole, just outside its event horizon. The fourth ray goes through the event horizon and spirals into the hole.

